The OSI Security Architecture

- The OSI security architecture has mainly three components.
  - Security Attack
  - Security Mechanism
  - Security Services

What is an Attack?

- Attack is an action that damages the security of an individual person or organization.
- When the attack is launched successfully then affects of the attack would be loss of the data or corruption of the data or ransomware attack or injection of the viruses or worms, or malicious software into the network or defacing the server or many affects are possible.
- These attacks are basically of two types
  - Passive Attack
  - Active Attack

Passive Attack:

- Passive attack is an attack that attempt to learn and make use of the information from the system.
- Passive attacks are un-authorized reading of message and no modification of messages involved.
- These passive attacks does not affect the system resources
- The main intention of these passive attacks are just **observing and eavesdropping or monitoring the transmission of the messages from sender to receiver.**

  Types of Passive Attack:

  1. Release of message content
     -Un-Authorized Reading of Messages
  2. Traffic Analysis
     - Locations, identity of the communicating host, length of messages and frequency of the messages that is being transmitted between sender and receiver.

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Active Attacks:

- ➔ Active attacks are involved in some modification of data stream or creation of some false stream.
- ➔ Active attacks are divided into four types.
  1. Masquerade attack
  2. Replay attack
  3. Modification of messages
  4. Denial of Service (DoS).

1. Masquerade Attack:

- ➔ In masquerade one person can pretends to be different persons.
- ➔ In this attack, the attacker can gain the someone credentials and log on to the system using those credentials and this attack would be a masquerading attack.

2. Reply Attack:

- ➔ In Replay attack, the attacker can send their own message on behalf of sender where the receiver can think about that messages are came from sender.

3. Modification of message:

- ➔ In this attack the attacker can modify the message and send it receiver.

4. Denial of Service (DoS):

- ➔ In this attack, when the user want to connect the server to access the service at that time the attacker can perform the some actions or activity on the server such that user is getting denied from accessing or getting the service.
- ➔ By this attack, the attacker overloading the server or putting lot of stress on the server such a way that the performance of server is degraded and server is no more able to serve the services to the new user.

Security Mechanism:

- ➔ Security Mechanism are of two types.
  1. Specific Security Mechanism
  2. Previous Security Mechanism

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1. **Specific Security Mechanism:**

➔ Specific security mechanism is incorporate with the protocol layers by default.

They are

- ❖ Encipherment
  - ➢ Encryption
  - ➢ Decryption
- ❖ Digital Signature
- ❖ Data Integrity
- ❖ Access Control
- ❖ Authentication
- ❖ Traffic Padding
- ❖ Routing Control etc.

Security Services:

➔ The processing or communication service that is provided by a system to give specific kind of protection to the system resources, security services implement security policies and are implemented by security mechanism.

➔ Some of the security services are

- ❖ Authentication
  - ➢ Peer to peer authentication
  - ➢ Data origin authentication
- ❖ Access Control
- ❖ Data Confidentiality
- ❖ Data Integrity
- ❖ Nonrepudation

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